

COSMOS DOCKET

Transmittal of complete written descriptions

ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY

Prepared 2026-09-07 for counsel. Not a USPTO filing. This preparer does not click Patent Center and does not sign as inventor.

Docket	Short title	Status	Exhibit
COSMOS-P01	MOTIF nine-stage loop (DEFINE first)	FILE	Ex. 01
COSMOS-P02	Dual-lane BUILD, no shared context	FILE	Ex. 02
COSMOS-P03	Family-axis occupancy; orthogonal Porosity tensor	FILE	Ex. 03
COSMOS-P04	Runtime-binding gate versus green-log	FILE	Ex. 04
COSMOS-P05	Adversarial occupancy engine (parent)	FILE (parent of products)	Ex. 05
COSMOS-P06	Local free-weight plurality versus S-tier	FILE	Ex. 06
COSMOS-P07	COSMOS Core: sole ledger, fence, tokens	FILE	Ex. 07
COSMOS-P08	DOM-first unmetered rail	FILE	Ex. 08
COSMOS-P09	Spend gate; confirm-to-widen	FILE	Ex. 09
COSMOS-P10	Resolver: sentinel root; existence is not identity	FILE (narrow)	Ex. 10
COSMOS-P11	SEED / signed session close (OPEN_CONTEXT)	FILE (narrow)	Ex. 11
COSMOS-P12	Crucible legal seats - HOLD	HOLD	Ex. 12
COSMOS-P13	ThinkFast drop-box daemon; ChatBot phone Freemium embodiment	FILE (twelfth \$65 slot; P12 remains HOLD)	Ex. 13

File each FILE packet as its own provisional (micro-entity \$65). P12 is HOLD. P13 takes the twelfth slot.

Attach HOW_IT_WORKS.pdf to each FILE at filing. Do not add it after the filing date.

Inventor legal name is blank. Counsel completes the cover sheet (PTO/SB/16) and micro-entity certification.

No claims are included. Counsel drafts claims. Paragraph numbers [0001] are for court and examiner citation.

Typography: letter, 1-inch margins, Times 12-point, 1.5 leading, line numbers, page n of m.

Do not post whitepaper, X, LinkedIn, or arXiv until FILE provisionals are in. Do not write V:\Ai from this stream.

COSMOS-P01

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Method for defining a feature as a frozen verbatim prompt and iterating a development loop by returning to definition then research**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P01**Kind:** Method**Status:** FILE**Fee class:** \$65 micro-entity provisional (37 CFR 1.16(d))**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister docket:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to computer-implemented methods for developing software with a plurality of generative models, and more particularly to a staged loop in which a feature is frozen as a verbatim prompt before research and in which iteration is required to re-enter that definition and then research.

BACKGROUND OF THE INVENTION

[0003] Prior multi-agent software-development loops commonly begin with a task statement that each agent is free to paraphrase. Iteration, when present, often returns to testing, critique, or planning, not to an external research stage with returns on disk. A critics-only polish cycle can improve the artifact already built; it cannot re-decide the feature when the wish has moved.

[0004] A frozen coder and a frozen architecture stop evolving. New models, new facts, and shifted needs never re-enter. The inventor observed that a missing DEFINE is a process scar: later critics cannot say whether the build is the thing that was decided, because nothing was frozen.

[0005] Related systems include staged multi-agent SDLC tools and planner-critic-executor pipelines. Those systems are identified below as information known to the applicant. They are not admitted to be prior art against any particular claim counsel may later draft.

BRIEF SUMMARY OF THE INVENTION

[0006] A nine-stage method is disclosed. Stage 1 DEFINE writes one clean prompt comprising what, why, acceptance, off-limits, and a live emit to honor. That file is the task text. Every subsequent model receives that exact text. No lane paraphrases it.

[0007] Stages 2 through 8 are RESEARCH (returns on disk before BUILD), ARCH, CONSENSUS (comparison and discussion), BUILD, CRITICS (comparison against DEFINE), CONSENSUS (adjudication), and IMPROVE (accept and apply; subtract as well as add).

[0008] Stage 9 ITERATE returns to stage 1 DEFINE, then to RESEARCH. It does not skip to CRITICS. Architecture, primary coder, and baselines may be re-decided. The gate of done is a value only the live system can emit (COSMOS-P04).

DEFINITIONS

[0009] As used herein, "DEFINE" means A frozen, written prompt that states the feature as what, why, acceptance, off-limits, and live emit, used verbatim by every subsequent model.

[0010] As used herein, "MOTIF" means The nine-stage development loop of this disclosure, encoded in the COSMOS tree as docs/MOTIF.md.

[0011] As used herein, "Lane" means An isolated builder attempt with private context.

[0012] As used herein, "Live emit" means A value only the running system can produce; see COSMOS-P04.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a flow diagram of the nine-stage MOTIF loop. Stage 1 is DEFINE. Stage 9 ITERATE returns to stage 1, then to RESEARCH, and does not return only to CRITICS.

[0014] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0015] Receive a work item that is not a one-line implement. If the item is a straight implement, the loop is not required.

[0016] DEFINE: write one clean prompt. Persist it as a file. That file is the work-order task text. Every later model (research, architecture, build, critic) is given those exact bytes. Do not paraphrase per lane. Do not start RESEARCH with an unfrozen vibe.

[0017] RESEARCH: obtain returns on disk before BUILD. Prefer rails that cannot run out of credit (COSMOS-P08). The same DEFINE text is the research question. Do not skip this stage.

[0018] ARCH: publish a decision rubric first. Independent designs. No peeking across designers.

[0019] CONSENSUS (comparison and discussion): converge, or mark CONTESTED with both positions and one line to the human disposer. No third model auto-resolves.

[0020] BUILD: dual-lane, no shared context (COSMOS-P02). Each spike must run. Same DEFINE text.

[0021] CRITICS: a different family from the builders. Judge against DEFINE (is this the thing we decided), not house style. Output is a comparison.

[0022] CONSENSUS (adjudication): reconcile critiques. Prior versions are baselines.

[0023] IMPROVE: accept and apply. Subtract as well as add. Net complexity and runtime cost should trend down as capability trends up.

[0024] ITERATE: return to DEFINE. If the wish moved, re-state the feature. Then RESEARCH. Run the whole cycle again (1 through 9), not 6 through 9.

[0025] Gate: runtime-binding (COSMOS-P04). Exit codes and green logs are not done.

[0026] Vendor plurality is required. Dual-lane BUILD and different-family CRITICS are seats of this loop (COSMOS-P02, COSMOS-P03, COSMOS-P05).

[0027] In a preferred embodiment the loop is driven by a native operating-system clock (approximately fifteen seconds) from a wishlist of directional input. The human names what and why. The mesh decides how. The loop is a process, not an endpoint.

[0028] Adversarial mechanics on the loop: DEFINE is the prompt; CRITICS is comparison; CONSENSUS is discussion and adjudication; IMPROVE is accept and apply.

BEST MODE

[0029] The best mode known to the inventor is the COSMOS encoding: docs/MOTIF.md (DEFINE is stage 1; ITERATE re-enters DEFINE), frozen DEFINE files under work_orders/ccr/, dual-lane BUILD through Gitur (GitHub, GitLab, Cursor), and the runtime-binding gate against live Core tree_id=KMesh-COSMOS-live.

FURTHER EMBODIMENTS

[0030] Embodiment A: a coding product (Forge) runs the loop on a named DEFINE file; critics include a Google family model and an OpenRouter value coder; the disposer is a single chief coder.

[0031] Embodiment B: the same loop is applied to a legal packet, a diligence packet, or an IP docket by changing the packet, not the loop (skins of COSMOS-P05).

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0032] keithbbf-gif/cosmos was created 2026-08-23T06:42:12Z. MOTIF.md is in the tracked tree. DEFINE-as-stage-1 was added 2026-09-07. United States inventor grace is about one year from the inventor's disclosure. Absolute-novelty countries may already be at risk for what GitHub enabled. Private-now does not un-publish.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0033] Related / crowding (information known to applicant, not an IDS, not a novelty opinion): ChatDev arXiv:2307.07924 (waterfall; iterate stays in test); MetaGPT arXiv:2308.00352; AutoGen arXiv:2308.08155; AgentCoder arXiv:2312.13010;

MapCoder arXiv:2405.11403 (debug reverts to planning); Twilio US20250165890A1 (planner-critic-executor); CN121436020A (eight-stage agent DevOps); CN121029145B; aiXplain US20260099419A1; Boehm spiral 1988; Scrum inspect-and-adapt.

[0034] This search did not find a United States patent whose claims require the last stage to re-enter a research stage with external returns on disk rather than test or critique. Unpublished applications are unknown. Official Patent Public Search remains owed.

WHAT THIS DISCLOSURE IS NOT

[0035] This disclosure is not any staged multi-agent SDLC. It is not ChatDev. It is not a patent filing. It is not a bake-off. It is not a claim that the inventor invented multi-agent software development.

STATEMENT OF INVENTION (NOT CLAIMS)

[0036] A development loop that defines the feature as a frozen verbatim prompt before research, and whose iterate step is required to re-enter DEFINE then research (fresh independent designs and new critics), rather than a critics-only polish cycle.

[0037] Counsel may draft claims. The foregoing is a statement of invention, not a claim set.

[0038] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0039] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P02

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Method of dual-lane building with a peeking ban and a single disposer**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P02**Kind:** Method / occupancy seating**Status:** FILE**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to seating two generative-model builders on the same decided work so that neither builder reads the other's context until compare, and so that only one disposer writes the live artifact tree.

BACKGROUND OF THE INVENTION

[0003] Shared-context debate lets a weak agent pollute a strong one. Builder-plus-nanny stacks a second seat hired to agree. OpenAI U.S. Patent No. 12,405,822 teaches a shared workspace - the anti-shape of this disclosure.

[0004] N-version programming and parallel contestants who later rank each other are related. They are identified below. This disclosure isolates two full builders, each of which must run, with a peeking ban until compare, plus a single pen.

BRIEF SUMMARY OF THE INVENTION

[0005] After consensus names the work, spawn two builder attempts. Each attempt has a private workspace and a private context. Neither builder may read the other's context, branch, or intermediate artifacts until compare. Each attempt must produce a running spike. Compare is a later consensus step, not a silent merge by a third model. The live tree has one writer. Builders propose; they do not hold the pen.

DEFINITIONS

[0006] As used herein, "Peeking ban" means A prohibition on a builder reading another builder's context, branch, or intermediate artifacts until a later compare step.

[0007] As used herein, "Disposer" means The single writer of the live tree. Also called chief coder or CCr in the source tree.

[0008] As used herein, "Propose" means To produce an artifact in a private workspace without writing the live tree.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 shows two builder lanes, each with a private workspace and private context, a later compare step, and a single disposer who alone writes the live tree. A dashed prohibition marks peeking.

[0010] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0011] After CONSENSUS names the work, spawn two builder attempts (Lane A and Lane B).

[0012] Give each attempt a private workspace and a private context. Do not share a transcript. Do not share a branch. Do not share intermediate artifacts.

[0013] Enforce the peeking ban until compare. A builder that reads the other lane is out of occupancy.

[0014] Each attempt must produce a running spike. A design that does not run is not a dual-lane result.

[0015] Compare is a later consensus step. Do not silently merge by a third model.

[0016] The live tree has one writer (COSMOS-P05). Builders propose. They do not hold the pen.

[0017] In a preferred embodiment Lane A is a Grok-class builder and Lane B is a Cursor-class builder on a branched tree. Gitur (GitHub, GitLab, Cursor) is the BUILD surface. The orchestrator does not author the live-tree implementation in-band.

[0018] This is two builders, not builder-plus-checker. A linter or critic seat is COSMOS-P01 CRITICS, not a second builder.

BEST MODE

[0019] Gitur dual-lane as operated 2026-09-04 and following: Lane A Grok 4.6 work-order session; Lane B Cursor Cloud Agent; same DEFINE text; no shared context; CCR disposes after review. Encoded docs/ADVERSARIAL_LOOP.md.

FURTHER EMBODIMENTS

[0020] Forge seats forge.ccr plus forge.adv_N are a product seating of the same occupancy. The chief-coder seat is locked to a named model in a preferred embodiment.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0021] Dual-lane is described in public cosmos MOTIF (repository created 2026-08-23). Gitur dual-lane (Grok plus Cursor) is operational occupancy, not a secret.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0022] Related: Galapagos N-version LLM arXiv:2408.09536; N-Version Programming with Coding Agents arXiv:2606.20158; Croto arXiv:2406.08979; IBM US20260252812A1; Avizienis 1985; Knight and Leveson 1986. Anti-shape: Twilio US20250165890A1 and AgentCoder (builder-checker); OpenAI U.S. 12,405,822 B1 (shared workspace); Aider Architect/Editor (executor sees the plan).

[0023] This search did not find a granted United States patent that forbids two coding agents from reading each other's context as the inventive occupancy. Unpublished applications are unknown.

WHAT THIS DISCLOSURE IS NOT

[0024] Not two agents coding in parallel in a shared chat. Not N-version at runtime with a voter. Not shared-ledger multi-agent. Not builder-plus-linter.

STATEMENT OF INVENTION (NOT CLAIMS)

[0025] Isolating two full builders (each must run) with a peeking ban until compare, plus a single disposer on the live tree.

[0026] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0027] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P03

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Method of family-axis occupancy using quantified porosity, pairwise orthogonal porosity vectors, and a tensor grid for seating generative models**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P03**Kind:** Method / occupancy rule**Status:** FILE**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to seating generative models for a task by family and by quantified mistake surface (porosity), including pairwise orthogonal porosity vectors on named axes and a tensor grid of those vectors, so that different-mistake models overlay residual holes faster than copies of the same family, at improved token efficiency.

BACKGROUND OF THE INVENTION

[0003] N copies of one API bill N times and keep the same holes. Self-consistency and more-agents-is-all-you-need stack copies. Hidden Clones (2026) and LLMs-as-Jury (2026) already measure family-correlated errors. Knight and Leveson 1986: independently written versions do not fail independently.

[0004] A single scalar quality score per model cannot tell an occupancy engine which other model punches a different hole. Two high-scoring copies of one family look interchangeable and still leave the same residual surface. Embedding cosine and representation orthogonality measure geometry of weights or activations, not observed task disagreement on the axes the operator cares about.

[0005] The inventor names the quality of a coder as porosity (spoken Pouroosity): the hole-set or mistake surface of that model (error classes, unseen tools, fabrication modes). Low-quality models with different mistakes, when overlaid, cover surface quickly at a greater-than-linear rate versus stacking copies. Copies of the same family leave holes aligned. This disclosure does not invent an exponent. Convert with a measurement. The measurement is a vector against another model, stored as a tensor, not a single value.

BRIEF SUMMARY OF THE INVENTION

[0006] Assign each model a family (training house, tool surface, observed failure mode). Quantify each coder's porosity as a hole-set measure and, pairwise, as a vector on named axes of interest, orthogonal to the other seated models. Pair magnitude equals disagreement frequency times error magnitude against the other model. More disagreement makes that pair vector more orthogonal. Persist observations in a measurement database and fold them into a tensor grid $T[\text{model } i, \text{model } j, \text{axis}]$. When seating votes, the same family does not count twice. Overlay: residual holes of the ensemble approximate the intersection of porosities when mistakes differ, and

approximate the first porosity when they align. Prefer adding a different-porosity model, including a low-quality one, over adding a high-quality copy of the same family. Use the tensor to maximize model choices for token efficiency and error discovery. Every adversarial trial in the occupancy engine (Forge and every profile) writes the tensor.

DEFINITIONS

[0007] As used herein, "Family" means An occupancy class: training house plus tool surface plus observed failure mode. Not a clustering theorem.

[0008] As used herein, "Porosity" means A hole-set measure of a coder (error classes / residual miss surface). Exact metric is an embodiment. Spoken Pourosity.

[0009] As used herein, "Different-mistake overlay" means Seating models whose hole-sets are not aligned, including a low-quality model of another family.

[0010] As used herein, "Orthogonal porosity vector" means The porosity of one model measured against another model on named axes of interest. It is a vector, not a scalar. Orthogonality of the pair rises with observed disagreement.

[0011] As used herein, "Pair magnitude" means The magnitude of an orthogonal porosity vector, equal to disagreement frequency times error magnitude for that pair. Unmeasured until both factors are observed.

[0012] As used herein, "Tensor grid" means The three-index array $T[i, j, a]$ of pair-vector components over models i, j and axes a , stored in a measurement database and used to seat models.

[0013] As used herein, "Axes of interest" means Named task axes on which disagreement and error are scored (for example coding, spec, security; law, facts; bull, bear, risk). A trial scores the axis it is running, not every axis by default.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a schematic of two hole-sets (porosities). Same-family copies leave holes aligned. A different-mistake overlay covers residual surface faster than stacking copies. Reason 1990 swiss-cheese is an analogy, not a claimed accident-causation theory. FIG. 2 is a pair-vector diagram: porosity of model i versus model j is a vector on named axes of interest, not a scalar. The magnitude of that pair vector equals disagreement frequency times error magnitude. The more the two models disagree, the more orthogonal the pair vector (aligned holes sit near-parallel). FIG. 3 is a tensor grid $T[i, j, a]$ stored in a measurement database and used to seat models so as to maximize error discovery per token.

[0015] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add

new matter after a filing date.

DETAILED DESCRIPTION

[0016] Assign each model a family. Occupancy rule, not a clustering theorem.

[0017] Quantify each coder's porosity $p(m)$ as a hole-set measure. Embodiments include pass/fail per task class and co-failure beta. Do not put a fake number in this specification.

[0018] Further quantify porosity as a pairwise vector. For models i and j and axis a , record whether an adversarial trial on a frozen task statement produced disagreeing ballots, and when known record the error magnitude of the residual hole (an embodiment uses a 1 to 10 scale). Pair magnitude on that axis equals disagreement frequency times mean observed error magnitude. Until error magnitude is observed, frequency may be known while magnitude remains unmeasured. Do not invent either factor.

[0019] Orthogonality: the more two models disagree on the axes of interest, the more orthogonal the pair vector. Models that keep the same holes (including same-family copies) sit near-parallel. This is an occupancy measurement of observed task disagreement, not a cosine of embedding vectors.

[0020] Persist each observation in a measurement database. An embodiment uses an append-only log as authority for the measurement and a rebuildable relational projection (for example SQLite) as cache. The operating-system ledger remains authority for jobs and spend; the porosity store does not replace it. A read that finds no observations reports unmeasured and does not create storage.

[0021] Fold observations into a tensor grid $T[i, j, a]$. Use the grid to seat models: prefer a candidate whose observed pair orthogonality versus already-seated models is large per token consumed. That is token efficiency together with error discovery. Unmeasured candidates sort last. Several inexpensive models chosen for disagreement can beat a single expensive model on price and on residual-hole cover.

[0022] When seating votes (architecture, build, critics): the same family does not count twice.

[0023] Overlay: residual holes of the ensemble approximate the intersection of porosities when mistakes differ; they approximate the first porosity when they align.

[0024] Prefer adding a different-porosity (different-mistake) model, including a low-quality one, over adding a high-quality copy of the same family. Predicted: surface cover grows faster than the same-family baseline (flat, or linear only in invoice).

[0025] Build the measurement into every adversarial trial. The coding profile (Forge) writes pair observations when two named models return on the same frozen statement. Every other occupancy profile (legal, medical, diligence, IP, site, physics) and every occupancy-engine trial (COSMOS-P05) uses the same hook. A silent rotating

free-model router is refused: a swap that looks like coverage is a hole.

[0026] Dual-lane (COSMOS-P02) and different-family critics (COSMOS-P05) exist to punch different holes, and they feed the tensor.

[0027] Hypothesis to measure, not a number in this provisional: coverage versus N for different-family overlays versus same-family copies. Same-family incremental coverage approximates zero. Different-mistake overlay: residual approximates the product of unaligned hole probabilities; cover of the remaining surface is large on the first different overlay. The tensor is how that hypothesis is converted with a packet rather than a fake exponent.

[0028] Swiss-cheese is an image (Reason 1990 as analogy). This disclosure does not claim Reason's accident-causation theory as the invention.

BEST MODE

[0029] Encoded docs/MOTIF.md and docs/arch/ORTHOGONAL_POROSITY.md: family-axis occupancy; pairwise orthogonal porosity vectors; JSONL observation log plus SQLite projection under the COSMOS runtime root; Forge facilitate and a shared trial hook; named pins rather than a rotating free-model router (silent swap is a hole that looks like coverage); dual-lane plus different-family critics.

FURTHER EMBODIMENTS

[0030] A coding critic seat uses a Google-family model and an OpenRouter value coder of another family, not two copies of one API. Their pair vector on the coding axis is written when their ballots differ. A later seating pass prefers a third inexpensive model whose tensor slice versus both is large per token.

[0031] Forge background free-coders on RESEARCH, ARCH, and CONSENSUS write the tensor from named :free pins. Crucible, Diligence, Differentiator, Docket, and Website GC trials call the same hook. Remote occupancy over a terminal or a phone (COSMOS-P05, COSMOS-P13) writes the same store.

[0032] A per-model scalar (errors per 100 lines times severity) may coexist as a fold. It does not replace the pair tensor.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0033] Swiss-cheese occupancy language is in MOTIF.md on public cosmos (method text 2026-08-25; swiss cheese 2026-09-07 still in-tree). Family-axis papers of 2026 are their measurements, not this inventor's. Pair-vector / tensor-grid language is inventor-named 2026-09-07 in the COSMOS tree before filing.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0034] Must name: Reason 1990 / BMJ 2000 (metaphor); Knight and Leveson 1986 and Brilliant, Knight and Leveson 1990 (N-version correlated failures); Kuncheva and Whitaker 2003 (ensemble diversity: Q-statistic, disagreement, double-fault); Jiang et al. ICLR 2022 (disagreement as a signal); Hidden Clones arXiv:2603.17111; CAPA / Great Models Think Alike arXiv:2502.04313; Nine Judges arXiv:2605.29800; LLMs-as-Jury arXiv:2607.10139; N-Version Programming with Coding Agents arXiv:2606.20158; co-failure ceiling arXiv:2606.27288; Wisdom and Delusion of LLM Ensembles arXiv:2510.21513; More Agents Is All You Need arXiv:2402.05120 (anti-shape: same-model copies); Self-consistency arXiv:2203.11171; Mixture-of-Agents arXiv:2406.04692; Self-MoA arXiv:2502.00674 (mixing can hurt); Estornell and Liu NeurIPS 2024; DZone/Cognaptus swiss-cheese AI as safety-stack metaphor; cosine similarity of embeddings or error-profile vectors as a contrast (this disclosure measures observed task disagreement on named axes, not embedding angle).

[0035] Patents on this point are thin. No United States patent found that claims same family is one vote, porosity overlay, pairwise orthogonal porosity vectors, or a disagreement-frequency times error-magnitude tensor grid for seating generative models. Unpublished applications are unknown. Official Patent Public Search completeness is unknown without a DOM rail.

WHAT THIS DISCLOSURE IS NOT

[0036] Not Reason's aviation model. Not ensembles exist. Not a measured superlinear exponent. Not a clustering theorem. Not cosine of embedding vectors as the invention. Not a fabricated score.

STATEMENT OF INVENTION (NOT CLAIMS)

[0037] Quantified porosity of each coder as a hole-set and as a pairwise orthogonal vector on named axes (magnitude = disagreement frequency times error magnitude), persisted as a tensor grid in a measurement database, with seating and overlay rules so that different-mistake (including low-quality) models cover residual surface faster than same-family copies, including token-efficient seating and a write from every adversarial trial.

[0038] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0039] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P04

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Method of binding completion to a live-system-emitted value with fail-closed refusal**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P04**Kind:** Method / selection gate**Status:** FILE**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to accepting a change to a running computer system only when the running system itself emits a value that only it can produce, and to refusing when that value is missing or contradicts the claim.

BACKGROUND OF THE INVENTION

[0003] A pipeline reports success because tests exited zero or a critic said it looks good. That is fabricated compliance: a plausible lie that closes the ticket while the running system does not do the thing. The inventor names this the placation class.

[0004] Continuous integration as a quality gate is related and, if treated as done, is the opposite occupancy.

BRIEF SUMMARY OF THE INVENTION

[0005] For a change to be accepted, require an artifact that only the running system can produce (a live API value, a ledger event field, a hash only the true run emits). Treat CI green, critic LGTM, and UI checks as projections, never authority. If the live emit is missing or contradicts the claim, record a refusal (fail-closed). Do not repair a green log in place. The report of the change must carry the live artifact or the refusal.

DEFINITIONS

[0006] As used herein, "Live emit" means A value only the running system can produce.

[0007] As used herein, "Projection" means A rebuildable view (dashboard, SQLite cache, CI log) that is never authority.

[0008] As used herein, "Fail-closed" means Missing or contradicting evidence is a refusal, not a silent pass.

[0009] As used herein, "Placation" means A plausible claim of compliance that the artifact contradicts.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 shows a claim of done entering a gate. A live emit from the running system passes. An exit code, a critic LGTM, and a dashboard color fail-closed as projections.

[0011] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0012] For a change to be accepted, require an artifact that only the running system can produce.

[0013] Examples of a live emit: an HTTP body from the resident Core bearing tree_id and a field only the true run can emit; a ledger event with a hash chain head; a response.model field that binds the model that actually answered.

[0014] Treat CI green, critic LGTM, and UI checks as projections, never authority.

[0015] If the live emit is missing or contradicts the claim, record a refusal. Do not repair a green log in place.

[0016] The report of the change must quote the live artifact or the refusal. Intention is not evidence.

[0017] A negative control may remain red on purpose. Painting it green is placation.

[0018] SCAR-derived law: every consequential action emits a machine-checkable artifact. Encoded docs/SCAR_PLACATION.md (2026-08-25) and kernel AD-11.

BEST MODE

[0019] COSMOS MOTIF gate against live Core at <http://127.0.0.1:8770>, tree_id=KMesh-COSMOS-live. Example: GET /jukebox returns 200 with QUEUED/RUNNING/BROKE/CLEAN counts; that body is the emit, not a test exit code.

FURTHER EMBODIMENTS

[0020] A model rail binds response.model, not the request. allow_fallbacks=false. A silent swap is a failed bind.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0021] Runtime-binding language is in public MOTIF and architecture (cosmos created 2026-08-23).

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0022] Related: Proof-or-Stop arXiv:2607.14890; Microsoft Azure 2026-08-29 validate at runtime; The New Stack runtime verification; CI as quality gate (Jenkins, GitHub Actions) as the opposite occupancy if CI is treated as done. No United States patent

found that claims only a live-system-emitted value is done. Patent occupancy thin. Paper and engineering occupancy high in 2026.

WHAT THIS DISCLOSURE IS NOT

[0023] Not don't-trust-CI as a slogan. Not a test framework. Not a bake-off.

STATEMENT OF INVENTION (NOT CLAIMS)

[0024] Binding finished to a value only the live tree can emit, with fail-closed refusal when that value is absent.

[0025] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0026] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P05

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Adversarial occupancy engine with isolated proposers, different-family critics, one disposer, and domain skins**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P05**Kind:** System / occupancy**Status:** FILE (parent of products)**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to an occupancy engine for generative-model work: isolated proposers, different-family critics versus what was decided, and one disposer, reused across domains by changing the packet rather than cloning the stack.

BACKGROUND OF THE INVENTION

[0003] If the orchestrator also holds the only wrench, the system ships a plausible lie. Shared-transcript crews put many agents in one context. Microsoft US20250371498A1 teaches the second agent based on the first - anti-isolation.

[0004] The inventor's new way is reliable, scalable AI: independent first, then argue; no shared context between builders; different-family critics; one pen. Products are skins, not new inventions.

BRIEF SUMMARY OF THE INVENTION

[0005] Name seats: disposer (one writer), N proposers, critics. Proposers run isolated; no shared transcript mid-pass. Critics are a different family from the builders; they judge the decision, not house style. Only the disposer writes the live tree or the grant tree. The same occupancy is reused across domains by changing the packet (code, casefile, deal room, IP docket), not by cloning the stack.

DEFINITIONS

[0006] As used herein, "Occupancy" means The seating of named models into named seats with isolation and one disposer.

[0007] As used herein, "Skin" means A product that reuses the engine by changing the packet and role names, not the occupancy.

[0008] As used herein, "Packet" means The work object: code change, legal casefile, diligence data room, IP docket, medical casefile.

[0009] As used herein, "Remote mouth" means A terminal (TUI / SSH / drop) or a phone (ChatBot / Voice) that cannot mount the live runtime root and that reaches the occupancy engine through the drop box of COSMOS-P13.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 shows seats: N isolated proposers, different-family critics, and one disposer writing the live tree. FIG. 2 shows the same engine with interchangeable packets (code, casefile, deal room, IP docket) as skins. FIG. 3 shows the same engine seated over remote: a terminal mouth or a phone mouth, neither mounting the live root, both dropping a typed work order (COSMOS-P13).

[0011] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0012] Name seats: disposer (one writer), N proposers, critics.

[0013] Proposers run isolated. No shared transcript mid-pass (COSMOS-P02).

[0014] Critics are a different family from the builders (COSMOS-P03). They judge the decision (COSMOS-P01 DEFINE), not house style.

[0015] Only the disposer writes the live tree or the grant tree. Everyone else proposes.

[0016] Reuse the occupancy across domains by changing the packet, not by cloning the stack.

[0017] MOTIF (COSMOS-P01) is the loop that runs this engine. Dual-lane (COSMOS-P02) is one seating.

[0018] Preferred skins: Forge (coding), Crucible (legal packet as occupancy not as a role-triad invention), Diligence (bull/bear/risk), Differentiator (anonymized medical casefile), Docket (IP). UPS is a physics skin that needs the inventor's July pack and is not invented here.

[0019] Fail-closed runtime-binding (COSMOS-P04) is the gate. The ledger (COSMOS-P07) is authority. Dashboards are projections.

[0020] Adversarial occupancy may be seated over remote. A terminal mouth or a phone mouth (the inventor: adversarial AI over remote, terminal or phone) drops a typed work order. The runner instantiates N isolated proposers and different-family critics. The disposer remains one. Neither mouth writes the live tree. This is not a shared-transcript crew on the phone and not a second Core.

[0021] Every adversarial trial writes pairwise orthogonal porosity observations into the measurement database of COSMOS-P03 (disagreement frequency times error magnitude on named axes; tensor grid for seating). Forge and every profile use the same hook. Unmeasured until observed.

BEST MODE

[0022] COSMOS Core as the OS; cDeck as a skin; products listed in docs/PROFILES.md; one CCR lease; orchestrator without the COSMOS pen. Encoded docs/CCR.md, docs/ADVERSARIAL_LOOP.md.

FURTHER EMBODIMENTS

[0023] Parallel native windows bound to {profile, core_base, tree_id} are two clients of one Core, not two Cores. A peer Core stays unnamed until the operator names a host.

[0024] Remote embodiment: adversarial AI over a terminal or a phone, same occupancy, same peeking ban, same one pen, ingress via COSMOS-P13.

[0025] Orthogonal porosity tensor (COSMOS-P03) is built into every trial this engine invokes, including remote mouths.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0026] CRUCIBLE is named in bts-mesh GitHub description 2026-08-16T10:21:14Z (word not in tracked files of that repo). Adversarial loop in public cosmos MOTIF, repository created 2026-08-23.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0027] Du 2023 debate; Irving 2018; Liang MAD; MoA; More Agents; Estornell and Liu; WO2025183627A1 (Lemon Inc MAD); US20240104125A1; US20250371498A1 (Microsoft, anti-isolation); AutoGen/ChatDev/MetaGPT/CAMEL/Magentic-One; Avizienis 1985; Knight and Leveson 1986; GitHub CODEOWNERS (VCS analog of one merger).

[0028] Combination of isolation plus family critics versus decided plus one disposer plus skins: unknown as blocking. Pieces crowded. This is not a novelty opinion.

WHAT THIS DISCLOSURE IS NOT

[0029] Not we invented multi-agent. Not Magentic-One (orchestrator holds the wrench). Not a voter over N-version binaries. Not a plaintiff-defense-judge patent (see COSMOS-P12 HOLD).

STATEMENT OF INVENTION (NOT CLAIMS)

[0030] One occupancy engine (isolate, different-family critique, one pen) reused as skins across coding, legal, medical, diligence, and IP packets, and reachable over remote from a terminal or a phone without mounting the live root, every trial writing the orthogonal porosity tensor of COSMOS-P03.

[0031] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0032] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P06

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Method of deploying isolated local free-weight models at fixed cost versus a metered S-tier pass**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P06**Kind:** Method / system of deployment**Status:** FILE**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to deploying two or three free-weight models of different families, isolated, on the same local machine, at low fixed cost, and comparing that occupancy to a single metered S-tier API pass.

BACKGROUND OF THE INVENTION

[0003] S-tier is a variable invoice (quota, key, billing). One brain, same porosity. Metered free APIs still run out. Local free weights plus isolation plus family-axis is how cost becomes a cap you can name.

[0004] Mixture-of-Agents, FrugalGPT, and blending papers block slogans. Microsoft U.S. 12,524,210 B2 (local GPT versus remote LLM for coding COGS, one local plus one remote) and Google US20240311405A1 (pick one of N models) must be read by counsel. Isolation (no shared context) is the thin element.

BRIEF SUMMARY OF THE INVENTION

[0005] Select N=2 or 3 models whose weights are free or open and whose families differ (COSMOS-P03). Deploy them on-box. Cost is machine plus power. Run them isolated (COSMOS-P02). Dispose through one writer (COSMOS-P05). Gate on a live emit (COSMOS-P04). Compare dollars and task outcome to a single S-tier API pass on the same task. Measurement is for conversion, not a number in this provisional.

DEFINITIONS

[0006] As used herein, "Free-weight" means Model weights that may be run locally without a per-token invoice from the weight provider.

[0007] As used herein, "S-tier" means A metered frontier API.

[0008] As used herein, "On-box" means On the operator's machine (llama.cpp / Ollama / vLLM class), not a hosted free API.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 shows two or three isolated local free-weight models on one machine (cost = hardware plus power) compared with one S-tier API (cost = tokens). A dashed cloud rail labeled OpenRouter :free is marked as a different rail, not this invention.

[0010] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0011] Select N=2 or 3 models whose weights are free or open and whose families differ.

[0012] Deploy them on-box. Cost equals machine plus power, not per-token.

[0013] Run them isolated: no shared context mid-pass.

[0014] Dispose through one writer. Gate on a live emit.

[0015] Compare dollars and task outcome to a single S-tier API pass on the same task. Do not invent a bake-off number in this specification.

[0016] OpenRouter free APIs are a different rail. The rotator is refused (silent model swap). A phone ChatBot named :free picker is that cloud Freemium free rail (COSMOS-P13 embodiment), not this packet. This packet is on-box weights - later actual unlimited on the user's hardware, and a Premium path of the ChatBot Freemium model.

[0017] Self-MoA teaches that mixing can hurt. This disclosure does not claim that mixing always helps.

BEST MODE

[0018] Two or three isolated local free-weight processes on one workstation, family-diverse, disposed by CCr, gated on a live emit, compared to one S-tier pass. Not yet a controlled bake-off.

FURTHER EMBODIMENTS

[0019] Gemma-class plus a different-family local coder plus an optional third family, each in its own process, no shared chat buffer.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0020] Swiss-cheese price/performance hypothesis in MOTIF.md and an unposted publish pack. Local multiplex as the inventor's 2026-09-07 measurement plan - not a public bake-off.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0021] Must cite: Mixture-of-Agents arXiv:2406.04692; Blending Is All You Need arXiv:2401.02994; FrugalGPT arXiv:2305.05176; RouteLLM arXiv:2406.18665; Hybrid LLM arXiv:2404.14618; More Agents arXiv:2402.05120; Self-MoA arXiv:2502.00674; LLM-Blender; PrivateGPT / GPT4All / LocalAI / OnPrem.LLM;

llama.cpp; Ollama; vLLM; Intelligence per Watt arXiv:2511.07885.

[0022] Patents counsel must read: US12524210B2 / WO2024238128A1 Microsoft; US20240311405A1 Google; Citibank US12536406B2; Martian US12314825B2; on-device multi-LM speculative decoding US12505335B2 (shares the decode stream - anti-isolation).

[0023] This search found no paper or patent that measures two or three isolated local free-weights versus one S-tier API on a coding task with box dollars versus token dollars. Unknown as blocking. Not a legal conclusion.

WHAT THIS DISCLOSURE IS NOT

[0024] Not run Ollama twice. Not FrugalGPT (paid API cascade). Not Together MoA (shared layers, hosted). Not a fake bake-off number. Not OpenRouter :free as unlimited.

STATEMENT OF INVENTION (NOT CLAIMS)

[0025] Family-diverse free-weight models, isolated, on one local machine at fixed cost, selected and gated as occupancy - versus one metered S-tier pass.

[0026] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0027] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P07

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

**Single-writer AI-work operating system with signed hash-chained ledger,
fencing tokens, and projections that are never authority****ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P07**Kind:** System**Status:** FILE**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree
discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click
USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35
U.S.C. 112(a).**Sister docket:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of
a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

*This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and
not a novelty opinion.*

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to an operating system for AI work whose authority is a single-writer append-only hash-chained service-signed ledger, with leases and fencing tokens, and with dashboards as rebuildable projections that are never authority.

BACKGROUND OF THE INVENTION

[0003] Two writers, empty-dir identity, advisory locks, and repair the log are measured failures. Green dashboards as authority are placation. Event sourcing, git, Kafka, and blockchains are related and are not this occupancy as an AI-work OS.

BRIEF SUMMARY OF THE INVENTION

[0004] One resident process is API, scheduler, arbiter, and ledger writer. The ledger is framed JSONL, hash-chained, service-signed. Replay rebuilds projections. Projections may not be written as authority. Workers receive a fencing token; the commit gateway rejects stale tokens and hash mismatches. A corrupt segment refuses; it is not repaired in place.

DEFINITIONS

[0005] As used herein, "Ledger" means Append-only, hash-chained, service-signed JSONL. Authority.

[0006] As used herein, "Projection" means A rebuildable view. Never authority.

[0007] As used herein, "Fencing token" means A monotonic token proving the holder still owns the lease.

[0008] As used herein, "Fenced commit gateway" means The only path by which a worker publishes; demands token plus expected input hashes.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 shows one resident service writing a hash-chained signed JSONL ledger. Workers present a fencing token at a fenced commit gateway. Dashboards and SQLite are projections. A corrupt segment refuses.

[0010] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0011] One process is API plus scheduler plus arbiter plus ledger writer. Fail-closed. Never a second unsynchronized writer.

[0012] Ledger: framed JSONL, hash chain, service signature. Replay rebuilds projections. Corrupt segment implies REFUSE plus incident, never repair-in-place.

[0013] Queue is immutable manifests plus ledger lifecycle events. SQLite is service-private projection only.

[0014] Workers run in attempt-private workspaces (native, DOM, cloud) and publish only through the fenced commit gateway presenting fencing token and expected input hashes.

[0015] Large artifacts live in a content-addressed store (filename equals hash; the ledger holds the live pointer).

[0016] The service is a modular monolith, split-ready: module interfaces are RPC-shaped so any module can later become a process without breaking the one versioned external API.

[0017] Scar-derived primitives are kernel interfaces. Workers cannot import around them (AD-11).

[0018] This OS is not MOTIF. MOTIF (COSMOS-P01) is how the OS builds the next organ. The house is the ledger, the fence, the resolver (COSMOS-P10), the seed (COSMOS-P11), the spend gate (COSMOS-P09), and the DOM rail (COSMOS-P08).

BEST MODE

[0019] COSMOS Core as ratified 2026-08-23, docs/FINAL_ARCHITECTURE.md, live tree_id=KMesh-COSMOS-live, serve on port 8770. Encoded cosmos/cosmos_service.py and related modules.

FURTHER EMBODIMENTS

[0020] Windows service recovery. Backup policy in Core; execution as scheduled jobs; rehearse-restore is first-class. Compatibility lane: legacy mutable-file tools serialized until they earn parallelism.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0021] Architecture text on cosmos created 2026-08-23T06:42:12Z. Predecessor mesh named on bts-mesh created 2026-08-16T10:21:14Z.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0022] Event sourcing (Fowler 2005); CQRS; git; Bitcoin; Kafka; US11943344B2 Ridgeline (hash-chained signed events plus projections, closest patent); WO2018217375A1 Microsoft signed log-chain; RFC 6962 Certificate Transparency; Kleppmann fencing tokens 2016; Chubby 2006; Raft 2014; etcd/k8s leases; SCSI-3 PR / STONITH; Temporal; Airflow; n8n; LangGraph.

[0023] Combination of all five as an AI-work OS: unknown as blocking. Not a novelty opinion.

WHAT THIS DISCLOSURE IS NOT

[0024] Not blockchain. Not git-as-authority. Not Temporal. Not event sourcing exists as a slogan.

STATEMENT OF INVENTION (NOT CLAIMS)

[0025] An AI-work OS whose authority is a single-writer signed hash-chained JSONL plus fencing-token commit of worker artifacts, with projections never authority.

[0026] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0027] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P08

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Scheduler routing that prefers a contained DOM rail because it cannot exhaust credit**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P08**Kind:** Method / routing policy**Status:** FILE**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to routing work in an AI operating system by preferring a browser-DOM rail because that rail does not depend on credit, quota, billing, key, or consent that can lapse, with metered APIs as explicit fallback.

BACKGROUND OF THE INVENTION

[0003] API-only agents die when the key, quota, or vendor dies. DOM is usually a tool, not the preferred unmetered path. Computer-use products are commonly metered.

BRIEF SUMMARY OF THE INVENTION

[0004] Register DOM workers as a scheduler rail, not a one-off script. Routing policy data: DOM first, API second; fallback is explicit and ledgery. Typed failures; report-never-retry unless contract-idempotent. Prefer this rail for RESEARCH because returns must not depend on prepaid remaining.

DEFINITIONS

[0005] As used herein, "DOM rail" means A contained browser worker with its own OS identity, ephemeral profile, and Job Object.

[0006] As used herein, "Typed failure" means One of UNREACHABLE, SESSION_EXPIRED, AUTH_REQUIRED, BROKE - not a generic exception.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 shows a scheduler. The preferred rail is a contained DOM worker. Metered APIs are an explicit audited fallback. Typed failures UNREACHABLE, SESSION_EXPIRED, AUTH_REQUIRED, BROKE are shown.

[0008] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0009] Register DOM workers as a scheduler rail, not a one-off script.

[0010] Contained workers: own OS identity, ephemeral ACL'd profiles, Job Objects.

[0011] Routing policy data: DOM first, API second. Fallback is explicit and audited on the ledger.

[0012] Typed failures: UNREACHABLE, SESSION_EXPIRED, AUTH_REQUIRED, BROKE. Report-never-retry unless the contract is idempotent.

[0013] Prefer this rail for RESEARCH (COSMOS-P01) because returns must not depend on prepaid remaining.

[0014] Dumping the DOM is not the rail. Interaction (evaluate, fill, send) is the rail.

[0015] A rotating free-model API is not this rail and is refused as a silent swap (COSMOS-P03 hole).

BEST MODE

[0016] COSMOS AD-6. Playwright-class contained workers. Live 2026-09-06 surfaces included Bing SERP, Cloudflare AI Playground, Copilot CLI. GEM via a named Chrome profile, never a banned URL. Ask the tools index before writing anything. Do not rebuild.

FURTHER EMBODIMENTS

[0017] Research returns written to disk from SGH and GEM in parallel (ROLD Rule 1). API is fallback.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0018] DOM-first in public architecture 2026-08-23.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0019] Playwright; Playwright MCP; browser-use; Stagehand; Skyvern; Magentic-One WebSurfer; US12101373B2 (browser RPA); CN119248379B (LLM plus Playwright scheduler); computer-use / Operator-class metered products. Distinctive prefer DOM because it cannot run out of credit as routing policy: unknown as patented. Ops folklore is widespread.

WHAT THIS DISCLOSURE IS NOT

[0020] Not browser automation exists. Not WICG scheduler.postTask. Not OpenAI Operator (metered).

STATEMENT OF INVENTION (NOT CLAIMS)

[0021] Scheduler routing that prefers a contained DOM rail because it cannot exhaust credit, with typed failures and audited API fallback.

[0022] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0023] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P09

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

**Fail-closed spend cap that cannot silently widen without typed confirm and
ledgered refusal****ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P09**Kind:** Method / system**Status:** FILE**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree
discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click
USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35
U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of
a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

*This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and
not a novelty opinion.*

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to gating spend on AI rails so that a cap cannot silently widen, a refused widen is itself recorded, and over-cap work does not run.

BACKGROUND OF THE INVENTION

[0003] OpenAI removed hard budget limits (notify-only). Admin APIs that raise max_budget without a two-step confirm. Silent fallback reroute that spends a different rail. Those are the scars.

BRIEF SUMMARY OF THE INVENTION

[0004] Each rail has a signed cap. A request that would raise the cap without a confirm token is refused (typed 409). The stored cap is unchanged. Widen requires an explicit confirm on the same cap object; ledger the grant or the refusal. Fail-closed: over-cap work does not run.

DEFINITIONS

[0005] As used herein, "Confirm token" means An explicit operator confirmation bound to the same cap object.

[0006] As used herein, "WIDEN_REQUIRES_CONFIRM" means The typed refusal when a widen is requested without confirm.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 shows a request that would raise a spend cap. Without a confirm token the gate returns a typed 409 WIDEN_REQUIRES_CONFIRM. The stored cap is unchanged. A signed ledger event records the refusal.

[0008] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0009] Each rail has a signed cap.

[0010] A request that would raise the cap without a confirm token is refused. Preferred embodiment: HTTP 409 WIDEN_REQUIRES_CONFIRM. The stored cap does not

move on the 409.

[0011] Widen requires an explicit confirm on the same cap object.

[0012] Ledger BUDGET_SET or SPEND_CAP_REFUSED. The refusal itself is evidence.

[0013] Fail-closed: over-cap work does not run.

[0014] Silent fallback to another rail that spends is also a widen and is refused unless confirmed.

BEST MODE

[0015] Measured in the COSMOS tree (docs/CHANGELOG_2026-08-30_CC_AUDIT.md). Signed BUDGET_SET / SPEND_CAP_REFUSED on the ledger.

FURTHER EMBODIMENTS

[0016] A POST to a budget route without confirm never moves the cap. A subsequent GET shows the old cap. That GET body is the live emit (COSMOS-P04).

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0017] Architecture and changelog in public cosmos tree (2026-08-23 and following).

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0018] LiteLLM virtual keys / tag budgets (admin can raise); Cloudflare AI Gateway spend limits 2026-06; Anthropic workspace limits; LangSmith evaluator weekly cap; US20250299128A1 multi-cloud budget throttle; EP4381451A1; PCI dual-control; 21 CFR 11.10. OpenAI hard-cap removal is the scar, not a teaching of confirm-to-widen. Combination (AI-rail cap plus no silent widen plus typed confirm plus signed ledger): unknown as blocking.

WHAT THIS DISCLOSURE IS NOT

[0019] Not AWS Budgets alerts. Not circuit breaker exists.

STATEMENT OF INVENTION (NOT CLAIMS)

[0020] An AI-rail spend cap that cannot move without a typed confirm, with the refusal itself ledgery.

[0021] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0022] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P10

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Boot identity of an AI runtime root by sentinel content where existence of a directory is not identity**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P10**Kind:** System (narrow)**Status:** FILE (narrow)**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to identifying a runtime root of an AI operating system by sentinel file content rather than by path existence, parent-walking, or a fallback ladder of drives.

BACKGROUND OF THE INVENTION

[0003] Empty-dir scar: a path that exists is treated as the install. Fallback ladders silently pick the wrong tree. Two writers on two roots. That is a deletion class when the wrong tree is overwritten.

BRIEF SUMMARY OF THE INVENTION

[0004] At boot, resolve one root by reading a sentinel file content, not by path existence. Match declared system and tree_id; refuse READY on mismatch or missing install record. Do not walk parents, do not try a list of drives, do not treat an empty directory as identity.

DEFINITIONS

[0005] As used herein, "Sentinel" means A file whose content (not its path) identifies the runtime root.

[0006] As used herein, "READY" means The service state that may serve. Forbidden without sentinel match.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 shows a boot resolver reading sentinel file content (.cosmos-root.json system and tree_id). An empty directory that exists is marked NOT IDENTITY. Parent-walk and drive-literal fallbacks are marked FORBIDDEN.

[0008] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0009] At boot, resolve one root by reading a sentinel file content, not by path existence.

[0010] Preferred sentinel: .cosmos-root.json declaring system=COSMOS and a tree_id. Preferred live tree_id: KMesh-COSMOS-live.

[0011] Match declared system and tree_id against the install record. Refuse READY on mismatch or missing install record.

[0012] Do not walk parents. Do not try a list of drives. Do not treat an empty directory as identity.

[0013] No import-time side effects. The resolver is instantiated at boot composition.

[0014] Two roots must not be conflated: the repo tree (tracked code) and the runtime root (live/). Existence of a folder is not identity of either.

[0015] Every path resolves through a declared role under that root. Nothing assembles a path by hand from a drive letter.

BEST MODE

[0016] cosmos_paths resolver. Service cannot go READY without sentinel-verified root. AD-5. Encoded cosmos/cosmos_paths.py.

FURTHER EMBODIMENTS

[0017] A unit test plants an empty directory at a tempting path and asserts the resolver refuses it.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0018] Resolver rules in public architecture 2026-08-23.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0019] OCI digest versus tag; Nix require-sigs / sandbox-fallback (fallback is the anti-pattern); git objects; chroot (path, not content); systemd-nspawn --root-hash; SLSA/in-toto/Cosign; HSTS fail-closed. No close patent found on JSON sentinel plus refuse empty-dir plus no parent-walk as AI-OS boot identity. Unknown as blocking. Narrow.

WHAT THIS DISCLOSURE IS NOT

[0020] Not chroot. Not Docker tags. Not a new hash algorithm.

STATEMENT OF INVENTION (NOT CLAIMS)

[0021] Boot identity of an AI runtime root by sentinel content, where existence of a directory is not identity and fallback ladders are forbidden.

[0022] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0023] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P11

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

Signed session close as a fail-closed operating-system incident**ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P11**Kind:** Method / system (narrow)**Status:** FILE (narrow)**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35 U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and not a novelty opinion.

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to making session identity of an AI operating system depend on a signed close manifest, such that a missing or invalid close is an incident and start refuses.

BACKGROUND OF THE INVENTION

[0003] Sessions vanish. The next session fabricates continuity. Vendor memory features store facts without a fail-closed close. CONTINUITY arXiv:2609.05269 (2026-09-05) is the closest paper and must be watched by counsel.

BRIEF SUMMARY OF THE INVENTION

[0004] close_session must emit a signed manifest covering declared fields (inherited facts, active leases, open watchers, handoff recipient). Absence, length mismatch, or HMAC failure is an incident (OPEN_CONTEXT), not a warning. start_session reads the manifest under its declared length and HMAC before injecting carry-over. An optional auto-resession clock consumes the same seed. A HOLD pause never self-clears.

DEFINITIONS

[0005] As used herein, "SEED" means A signed context manifest written at close (SEED.json plus declared length/HMAC).

[0006] As used herein, "OPEN_CONTEXT" means The incident raised when close lacks a valid manifest.

[0007] As used herein, "HOLD" means A pause that never self-clears.

[0008] As used herein, "Resume gate" means A pause at boot that may auto-resume on a native clock if the operator does not choose.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 shows close_session emitting a signed SEED.json (declared length and HMAC). Absence is OPEN_CONTEXT. start_session refuses NO_SEED, BAD_SEED, and IDENTITY_MISMATCH. A HOLD pause never self-clears; a resume-gate pause may auto-resume on a native clock.

[0010] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add

new matter after a filing date.

DETAILED DESCRIPTION

[0011] close_session must emit a signed manifest covering declared fields: inherited facts, active leases, open watchers, handoff recipient.

[0012] Absence, length mismatch, or HMAC failure is an incident (OPEN_CONTEXT), not a warning.

[0013] start_session reads the manifest under its declared length and HMAC before injecting carry-over. It refuses NO_SEED, BAD_SEED, and IDENTITY_MISMATCH.

[0014] A lightweight session pointer may sit beside the seed (BUCm.toml in the source tree) as one truth, never a competing second handoff.

[0015] Resume gate: after boot loads carry-over, one option (resume all, subset, hold). No affirmative selection implies auto-resume on a native clock. Default is motion.

[0016] A HOLD never self-clears. The two pause kinds are not the same.

[0017] Carry-over is why the OS survives context death. MOTIF is what the clock drives after the seed is accepted.

[0018] This is not ChatGPT memory.

BEST MODE

[0019] cosmos_session.close_session / start_session. AD-10. docs/PAUSE_PROTOCOL.md. Native fifteen-second clock (cosmos_watchdog2.py) honors mode in the pause flag.

FURTHER EMBODIMENTS

[0020] Auto-resession watermark at about seventy percent of context; hard close at ninety percent; pack to live/state/session_saves/. Improper close: resume from session log and/or running_session_file.toml.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0021] AD-10 in public architecture 2026-08-23.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0022] CONTINUITY arXiv:2609.05269 (2026-09-05 - signed context manifests plus CLOSE/OPEN receipts; closest paper; watch); Portable Agent Memory arXiv:2605.11032; Context Passport; CRAFT handoff; ctx handover; vendor memory; CRIU; JWT/JWS; US20250259069A1 reconstitutable sessions. Incident-on-missing-close as OS rule: unknown as patented.

WHAT THIS DISCLOSURE IS NOT

[0023] Not ChatGPT memory. Not CRIU. Not write a notes file.

STATEMENT OF INVENTION (NOT CLAIMS)

[0024] Session identity that must close with a signed manifest; missing close is an incident and start refuses.

[0025] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0026] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P12

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

**Crucible as a skin of occupancy applied to a legal packet (HOLD - do not
file as plaintiff-defense-judge)****HOLD — DO NOT FILE AS A PLAINTIFF-DEFENSE-JUDGE INVENTION****Docket:** COSMOS-P12**Kind:** Method / application of COSMOS-P05**Status:** HOLD**Fee class:** \$65 only if counsel files occupancy-not-roles**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree
discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click
USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35
U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of
a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

*This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and
not a novelty opinion.*

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure, if filed, relates to applying the occupancy engine of COSMOS-P05 to a legal packet. It does not relate to inventing an AI plaintiff, defense, and judge.

BACKGROUND OF THE INVENTION

[0003] The plaintiff-defense-judge triad is crowded: SimuCourt/AgentsCourt 2024, AgentCourt 2024, CN119168059B granted 2025-07-15, US20260037351A1 claim 21. File only as occupancy applied to a legal packet. Roles are embodiments.

[0004] Public bts-mesh GitHub description (2026-08-16) says CRUCIBLE method. Tracked files in that repo: zero hits for the word CRUCIBLE. The public date is a name, not a method disclosure in code.

BRIEF SUMMARY OF THE INVENTION

[0005] HOLD. Do not claim AI plaintiff, defense, and judge. If counsel files, the specification is P05 occupancy applied to a legal packet: isolate, different-family, one disposer, runtime-bind, ledger. Role names are embodiments.

DEFINITIONS

[0006] As used herein, "Crucible" means A product skin of COSMOS-P05 applied to a legal packet. Not a TM clearance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 (if filed) shows P05 occupancy applied to a legal packet. Role names plaintiff, defense, and judge are labeled EMBODIMENT, not independent invention. A HOLD legend marks this packet not for filing as a role-triad.

[0008] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0009] Do not claim AI plaintiff, defense, and judge as the invention.

[0010] If filed: apply COSMOS-P05 occupancy to a legal packet. Isolated proposers. Different-family critics. One disposer. Runtime-binding gate. Ledger authority.

[0011] Role names (plaintiff, defense, judge) are embodiments of seats, not independent claims.

[0012] Spend-gated critics. Vendor-plural.

[0013] Marks: CrucibleTech RN 8043152 (games); Star Lab CRUCIBLE RN 5023199 (OS); Crucible Discovery; Destiny PvP fame. Not a TESS clearance.

[0014] The twelfth sixty-five-dollar slot is taken by COSMOS-P13. This packet stays HOLD unless counsel files occupancy-not-roles as an extra application.

BEST MODE

[0015] In-tree POST /api/v1/crucible. 501 if no critics. Own tree per occupant. Product profile docs/PROFILES.md. Do not mix with medical Differentiator or Diligence packets.

FURTHER EMBODIMENTS

[0016] A federated sit: a Crucible window bound to another operator's Core when that operator has a host. Until a host is named, peer is NO_HOST.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0017] Name CRUCIBLE method on GitHub About 2026-08-16T10:21:14Z. United States grace for what that string disclosed is thin (a name, not a spec).

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0018] BLOCKING for role-triad claims: SimuCourt/AgentsCourt arXiv:2403.02959; AgentCourt arXiv:2408.08089; LegalSim; SAMVAD; PROCLAIM arXiv:2603.28488; 3-Ply arXiv:2606.30906; CN119168059B granted; US20260037351A1; US20250148558A1 (jury-side). Products: Jury Simulator, Juries.ai, Trial AI.

WHAT THIS DISCLOSURE IS NOT

[0019] Not we invented mock trial AI. Not a TM clearance. Not a twelfth slot while P13 is FILE.

STATEMENT OF INVENTION (NOT CLAIMS)

[0020] If filed: Crucible as a skin of P05, not a PDJ patent. Else fold into P05.

[0021] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0022] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

COSMOS-P13

UNITED STATES PATENT AND TRADEMARK OFFICE

Provisional Application for Patent — Written Description

**Voice or chat client drop of a typed work order onto a reachable inbox
executed by a host OS daemon that instantiates a named agent and audits
by timestamps and a signed ledger****ATTORNEY WORK PRODUCT — NOT A FILED PATENT APPLICATION — FOR COUNSEL ONLY****Docket:** COSMOS-P13**Kind:** System / method**Status:** FILE (twelfth \$65 slot; P12 remains HOLD)**Fee class:** \$65 micro-entity provisional**Date of this written description:** 2026-09-07**Name of inventor:** _____ (counsel to complete). The source tree
discloses the operator as Keith. This preparer does not sign as inventor.**Preparer:** COSMOS Chief Coder session (not the inventor). Does not sign as inventor. Does not click
USPTO.**Claims:** none herein. Counsel drafts claims. Paragraphs below are a written description under 35
U.S.C. 112(a).**Sister dockets:** COSMOS-P01 through COSMOS-P13. This provisional does **not** claim the benefit of
a sister. Duplicate HOW_IT_WORKS.pdf into this filing.

*This document is attorney work product prepared for counsel. It is not a filed application, not legal advice, and
not a novelty opinion.*

CROSS-REFERENCE TO RELATED DISCLOSURES

[0001] COSMOS-P01 through COSMOS-P13 are related written descriptions prepared the same day. A provisional cannot claim the benefit of a sister provisional. The how-it-works appendix is incorporated as if fully set forth and must be attached at filing. Do not add technical matter after a filing date (37 CFR 1.53(c)).

FIELD OF THE INVENTION

[0002] The present disclosure relates to ingress of work from a voice, chat, phone, or remote-terminal client that cannot see an authority runtime root, via a typed work order on a reachable repository folder, executed by a host operating-system daemon that creates a named agent or the adversarial occupancy engine of COSMOS-P05, collects output, and writes a timestamped audit, without deleting the repository object and without letting the client write the live tree.

BACKGROUND OF THE INVENTION

[0003] If the human is the wire, voice work dies when the chat dies. If GitHub Actions runs the job, the audit is a vendor log and the agent is not fenced. If the phone could write the live root, that is a second writer (two-writer deletion scar).

[0004] The inventor observed that the combination - client cannot mount live; typed JSON on GitHub; host OS daemon not Actions; named agent; done equals output file exists; timestamps plus signed ledger; never delete the drop - was not found as a blocking United States claim set. Pieces are crowded. This is not a novelty opinion. Patent Public Search remains owed.

BRIEF SUMMARY OF THE INVENTION

[0005] An operator (voice, Chatbox, phone, or remote terminal) who cannot see the live runtime root writes a typed JSON work order to a GitHub folder that they can reach. A native system daemon on the host monitors that drop box, files the order into the live bucket, creates a session of the named agent or seats the adversarial occupancy engine (N isolated proposers, different-family critics, one disposer), collects Output, and records results with timestamps and an audit trail. GitHub files are never deleted. The LLM does not poll itself. GitHub Actions is not the executor. Dispose of live-tree writes remains one chief coder. A consumer ChatBot phone is an embodiment mouth: named OpenRouter :free picker, Freemium (free forever phone, free Desktop install, optional Premium). Adversarial AI over remote (terminal or phone) is an embodiment of COSMOS-P05 through this ingress, not a fourteenth provisional.

DEFINITIONS

[0006] As used herein, "Drop box" means A GitHub path work_orders/drop/ on a named repo and branch. Not the live tree. Not repo root.

[0007] As used herein, "Six fields" means Agent; Context source; Task; Target and scope; Timestamp; Output.

[0008] As used herein, "WRITE-PRIVATE" means The agent may write only its Output folder or filename.

[0009] As used herein, "Freemium" means Free forever usable phone chat on named :free models; free Desktop install; optional Premium. Not a trial. Not crippleware. Not a claim that a vendor free API is infinite.

[0010] As used herein, "Remote mouth" means A terminal (TUI / SSH / drop) or a phone (ChatBot / Voice) that reaches this ingress without mounting the live root.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 shows a client that cannot mount the live root writing a typed JSON work order to a GitHub folder. A native OS daemon lists the drop, files DROPPED, instantiates a named agent WRITE-PRIVATE to Output, and records timestamps. GitHub objects are never deleted. GitHub Actions is not the executor. FIG. 2 shows a consumer ChatBot phone as a mouth: named :free model picker, Freemium free forever, funnel to Desktop ChatBot, optional Premium. FIG. 3 shows two remote mouths - a terminal and a phone - either of which may seat the adversarial occupancy engine (COSMOS-P05) through the same drop box.

[0012] The drawings are described in prose so that a person of ordinary skill can produce sheet drawings. Sheet drawings may be added by counsel before filing. Do not add new matter after a filing date.

DETAILED DESCRIPTION

[0013] Inbox the client can reach equals GitHub path work_orders/drop/ on a named repo and branch. Not the live tree. Not repo root. One JSON object, six fields. Filename Windows-legal.

[0014] A native OS clock (Windows Scheduled Task plus a pythonw daemon in the preferred embodiment) lists the drop, parses, and files DROPPED into the live bucket. It does not execute the Task.

[0015] Seen-set is append-only (sha). GitHub objects stay. Seen-set is not authority.

[0016] A second native runner picks DROPPED, creates a session of the named Agent, passes Task plus read-only context, WRITE-PRIVATE to Output only.

[0017] Collect: DONE if and only if the Output file exists; else FAILED. File the order into the assigned or done folder with timestamps.

[0018] Audit: daemon heartbeats (last_run_epoch), ISO timestamps on the order, ledger events when Core is composed. Dashboards are projections (COSMOS-P07).

[0019] Voice speech-to-speech (Grok Voice Think Fast 2.0 in the operator embodiment) is one mouth. Chatbox is the same inbox. Return path may be a drive the phone can read without mounting live/.

[0020] Dispose of live-tree writes remains one chief coder. The daemon does not hold the pen (COSMOS-P05).

[0021] Consumer embodiment - ChatBot phone: the mouth runs on OpenRouter named :free models (low latency). The human picks a named model. Rotating ids openrouter/free and openrouter/auto are refused. Runtime bind equals response.model (COSMOS-P04).

[0022] Freemium: Free forever is real chat on named :free models, no trial clock, no invented quota from the operator. Install (free) is Desktop ChatBot. Premium is optional (named paid models, desktop hands, later COSMOS-P06 on-box weights). Price is named by the operator, not in this specification.

[0023] Vendor :free caps remain (in the preferred OpenRouter embodiment, twenty requests per minute; fifty per day until ten dollars lifetime credits, then one thousand per day). Do not sell unlimited :free as Premium. Do not print no usage limit as a warranty of vendor capacity.

[0024] The phone never mounts live/. A remote terminal is a mouth, not a second writer on the runtime root. cosmos-voice.apk remains a draft seed, not this product. This is not CVM. This is not a thirteenth provisional.

[0025] Adversarial over remote (the inventor: terminal or phone): the Agent field may name the occupancy engine (a seat-set) rather than one model. The runner then creates N isolated sessions under the peeking ban (COSMOS-P02). Critics are a different family. One disposer still writes the live tree. Free-tier single-model chat on the phone remains usable; adversarial N-seat is occupancy through the same mouths, not a shared chat on the device.

BEST MODE

[0026] Measured loop: Think Fast 2 (phone) to keithbbf-gif/cosmos work_orders/drop/*.json to ingest clock (about fifteen seconds, schtask COSMOS SGH Drop Ingest, cosmos_sgh_drop_ingest.py) to live/state/work_orders/bucket/ to Work-Order Runner creating an Agent session WRITE-PRIVATE; Output exists equals DONE; assigned folder plus heartbeat JSON plus ISO timestamps. CCr --accept / --reject remains the only live-tree writer. DEFINE files: DEFINE_THINKFAST_DROP.md and DEFINE_CHATBOT_PHONE.md.

FURTHER EMBODIMENTS

[0027] Operator mouth: SGH Android Voice plus Grok Voice Think Fast 2.0 (tabled as voice refine; still the operator path).

[0028] Consumer mouth: ChatBot phone Freemium as defined above. Desktop binary named at build (OpenWork desktop versus a COSMOS desktop ChatBot). Do not iframe OpenWork Web. Do not charge OpenWork Web fifty dollars as this funnel.

[0029] Remote terminal mouth: a TUI or SSH session that can write GitHub and cannot mount live/. Same six-field drop. Same daemon. May seat one agent or the adversarial engine.

DISCLOSURE CLOCK (ALREADY PUBLIC)

[0030] Work-order SOP and work_orders/drop/ exist on public cosmos after 2026-08-23. Voice loop named in wishlist and routing. GitHub date for the method as a voice-to-daemon-to-agent-to-audit loop is thinner than Core's 2026-08-23 architecture dump. FILE before another public article.

INFORMATION CONCERNING RELATED ART (NOT AN IDS; NOT A NOVELTY OPINION)

[0031] GitHub Issues / PR templates as inbox; GitHub Actions on push; ChatOps (Slack to Jenkins); cron plus file drop; n8n / Zapier / Make webhooks; Temporal / Airflow / Luigi; Siri Shortcuts / Alexa skills calling HTTP; Twilio US20250165890A1; voice-to-ticket SaaS.

[0032] Combination not found as a blocking United States claim set: voice or chat client that cannot mount the authority root drops a typed work order on GitHub; a host OS daemon (not Actions) monitors, instantiates a named AI agent, collects a single Output file as DONE, and writes a timestamped COSMOS audit while never deleting the GitHub drop and never letting the agent write the live tree. Unpublished applications unknown. DOM research of Patent Public Search still owed.

WHAT THIS DISCLOSURE IS NOT

[0033] Not GitHub Actions. Not a mobile backend. Not a mobile frontend. Not CVM. Not we invented work queues. Not cron. Not the MOTIF loop (COSMOS-P01). Not the ledger primitive (COSMOS-P07). Not a claim that OpenRouter :free has no rate limit. Not a thirteenth sixty-five-dollar slot.

STATEMENT OF INVENTION (NOT CLAIMS)

[0034] A voice-reachable or terminal-reachable GitHub drop box, executed by a local OS daemon that creates a fenced agent or seats the adversarial occupancy engine, with DONE equal to Output file and authority equal to timestamps plus signed ledger, operator out of the execution wire; optionally a Freemium phone ChatBot mouth on named free models that funnels to a desktop install.

[0035] Counsel may draft claims. The foregoing is a statement of invention, not a claim set under 35 U.S.C. 112(b).

APPENDIX TO ATTACH AT FILING

[0036] HOW_IT_WORKS.pdf (APP_OS: SCAR, ROLD, carry-over; APP_COSMOS; APP_CRUCIBLE; APP_BTS_MESH). Duplicate the appendix into this provisional at filing. A provisional cannot claim benefit of a sister.

How BTS-MESH, CRUCIBLE, and COSMOS work

OS first: SCAR, ROLD, carry-over. MOTIF is how it builds.

Written-description appendices for US provisionals

Not a USPTO filing. Not legal advice. Attach these PDFs to the provisional packets at filing. Do not add them after a filing date. Keith + Legal file. This TUI does not click USPTO.

September 2026

Appendix - COSMOS is the OS (not MOTIF)

Kind: written description for Legal. Not a USPTO filing. **Keith 2026-09-07:** *The SCAR system, the ROLD, the CARRY OVER SYSTEM - COSMOS is much more than just MOTIF.*

MOTIF (P01) is the **method that builds** the house. COSMOS is the **house**: an operating system whose law is earned from measured failures, whose desk of record is ROLD, and whose sessions **carry over** as a structural close/open, not as chat memory.

Three organs MOTIF is not

1. The SCAR system

A **scar** is a failure that was **measured**, then encoded as law. ROLD SCARS.md (local v:\Ai\ROLD, GrokBot tree): append-only, dated, numbered, **never edited**. Wrong entries stay; a later correction names the entry it corrects. That is ADR/incident class (Nygard): context to what happened to consequence.

COSMOS turns scars into **kernel interfaces** (AD-11): workers cannot import around them. The worst class is **placation** (docs/SCAR_PLACATION.md, 2026-08-25): a plausible claim of compliance that the artifact contradicts. Guard: every consequential action emits a machine-checkable artifact; a report **quotes the artifact**, never the intention; missing or contradicting artifact is a refusal, surfaced first. Runtime-binding (P04) is this guard as a gate.

Empty-dir identity, two-writer deletion, green-log DONE, bash-mount CRLF phantoms - each is a scar that became a primitive (resolver, one pen, live emit, host-side ground truth). MOTIF *uses* those primitives. It did not invent them.

2. The ROLD

ROLD = the desk of law / repository of documents. Keith 2026-07-30: a repository, not a flat dump. Organising rule: **a document is defined by what makes it change**. If two things in one file change for different reasons, they are two files.

Always-granted across streams: rules, glossary, scars, rails, routines. **Ask the tools index first. Do not rebuild.** ROLD Rule 1: SGH and GEM first, both, in parallel, on DOM. The index is identity of capability; existence of a folder is not a tool.

ROLD is not MOTIF stage 2 RESEARCH. RESEARCH *reads* ROLD. The desk exists whether or not a MOTIF tick is running.

3. The carry-over system

Close of a session **must** write a signed context manifest (SEED.json, declared length/HMAC): inherited facts, active leases, open watchers, handoff recipient. Close without it is **OPEN_CONTEXT**. Start refuses NO_SEED / BAD_SEED / IDENTITY_MISMATCH (P11, AD-10).

BUCm.toml is the lightweight session pointer beside the SEED - one truth, never a competing second handoff.

Resume gate (all streams): after BootUP loads carry-over, one option (resume all / subset / hold). No affirmative selection to **AUTO-RESUME** on the native 15s clock. Default is MOTION. A **HOLD** never self-clears. Two pause kinds are not the same.

Carry-over is why COSMOS survives context death. MOTIF is what the clock drives **after** the seed is accepted.

How they sit together

Organ	Job	Packet
SCAR system	Law earned from measured failure; anti-placation	P04 + kernel AD-11

Organ	Job	Packet
ROLD	Desk of record; ask before rebuild	supporting disclosure (this appendix)
Carry-over	Signed close; resume gate; auto-resession	P11
Core	One writer, ledger, fence	P07, P10
Ingress	Voice/Chatbox to GitHub drop to daemon to agent to audit	P13
MOTIF	How the OS builds the next organ	P01 (DEFINE first)

File the OS. File MOTIF. Do not file MOTIF as if it were the OS.

Attach to

P04, P07, P10, P11, P13, and the how-it-works pack. Duplicate at filing.

Appendix - How COSMOS works

Kind: written description / drawings-in-prose for Legal to attach to US provisionals. **Status:** not a USPTO filing. Not legal advice. Not a novelty opinion. **Source of truth:** docs/FINAL_ARCHITECTURE.md ratified 2026-08-23 (docs/RATIFIED.md); live Core tree_id=KMesh-COSMOS-live on this machine. **Maps to packets:** P07 (ledger + fence), P10 (resolver), P11 (SEED), P08 (DOM-first), P09 (spend). MOTIF (P01) is how the OS builds the next organ - not the OS. See **Appendix - COSMOS is the OS** (SCAR, ROLD, carry-over).

One paragraph (the invention as an OS)

One resident Windows service - **COSMOS Core** - is the sole authority: API gateway, scheduler, lease arbiter with fencing tokens, spend gate, return-watcher, registry + prober, and the single writer of an **append-only, hash-chained, service-signed JSONL ledger** on a verified native volume. Everything else - queue views, registries, dashboards (KDash / cDeck), spend totals - is a **rebuildable projection** (service-private SQLite allowed as cache, never authority). Large artifacts live in a **content-addressed store** (filename = hash; the ledger holds the live pointer). Workers (native, DOM, cloud) run in attempt-private workspaces and publish only through a **fenced commit gateway** that demands a fencing token and expected input hashes. The service is a **modular monolith, split-ready**: module interfaces are RPC-shaped so any module can later become a process without breaking the one versioned external API that cDeck, voice, and other clients consume.

COSMOS is the **operating system**. Products (Forge, Crucible, Diligence, Docket, UPS, Differentiator) are **skins** on that OS, not second Cores.

What a later claim must be able to point at

A person of ordinary skill, given this appendix plus the live tree, can build:

- 1 A single process that is API + scheduler + arbiter + ledger writer.
- 2 A framed JSONL ledger with hash chain and service signature; replay rebuilds projections; a corrupt segment **REFUSES** rather than repairing in place.
- 3 A boot resolver that takes **one** install-configured root, verified by sentinel **content** - existence of an empty directory is not identity.
- 4 Workers that cannot write the live tree except through the fenced commit gateway.
- 5 One versioned HTTP API (/api/v1/. . .) as the only external surface.
- 6 DOM as a first-class rail (unmetered path) with API as fallback, both audited.
- 7 A spend gate that will not silently widen a cap.
- 8 A signed session-close manifest (SEED) so carry-over is structural.

Decisions (ratified 2026-08-23)

#	Decision
1	One resident service. Fail-closed. Never a second unsynchronized writer.
2	Leases + monotonic fencing tokens + fenced commit gateway. Advisory locking is dead.
3	Ledger = framed, hash-chained, signed JSONL. Corrupt segment ⇒ REFUSE + incident.
4	Queue = immutable manifests + ledger lifecycle events. SQLite is projection only.
5	Resolver instantiated at boot; service cannot go READY without sentinel-verified root.
6	DOM is a first-class scheduler rail (contained workers, typed failures).
7	One versioned API. UI may deploy separately; authority may not.
8	Backup: policy/verification in Core; execution as scheduled jobs; rehearse-restore is first-class.
9	Compatibility lane: legacy mutable-file tools SERIALIZED until they earn parallelism.
10	Context manifests at session close. Closure without a valid manifest ⇒ OPEN_CONTEXT.
11	Scar-derived primitives are kernel interfaces; workers cannot import around them.

Two roots (do not conflate)

- **Repo tree** - tracked code and governance (cosmos/, docs/, tests/).
- **Runtime root** - the running instance (live/: state, ledger, queue, registry, config).

Identified by `.cosmos-root.json` (system=COSMOS, tree_id=KMesh-COSMOS-live).

The resolver takes one configured root. No drive literal as identity, no parent-walking fallback ladder, no import-time side effects.

Method that builds the OS (MOTIF)

COSMOS builds itself on directional input. The loop is **nine** stages: **DEFINE** to RESEARCH to ARCH to CONSENSUS to BUILD to CRITICS to CONSENSUS to IMPROVE to ITERATE (back to **DEFINE**, then RESEARCH). Dual-lane BUILD: two builders, **no shared context**, different families. Critics judge *is this the thing we decided*, not house style. One disposer (CCr) writes the live tree. The runtime-binding gate is a value **only the live tree can emit** - never an exit code, never a green log.

What this is not

Not a second blockchain. Not git-as-authority. Not “event sourcing exists.” Not a dashboard that is itself the OS. Not OpenWork (orch harness) and not cDeck (skin). cDeck is a client of `/api/v1`. OpenWork is a separate app/window.

Already public (clock)

GitHub cosmos created 2026-08-23T06:42:12Z. Ratification record that day. Predecessor mesh named on GitHub `bts-mesh` 2026-08-16. July 2026 operation is **local BTS-MESH run records**, not that GitHub date. See appendix How BTS-MESH works.

Attach to

FILE packets P01, P04, P07, P08, P09, P10, P11 at minimum. Duplicate this PDF into each provisional so each filing is enabling on its own (a provisional cannot claim benefit of a sister provisional).

Appendix - How CRUCIBLE works

Kind: written description for Legal to attach to US provisionals. **Status:** not a USPTO filing. **P12 (role-triad as the invention) remains HOLD.** This appendix describes the **occupancy engine applied to a legal packet** (a skin of P05), not “AI plaintiff / defense / judge” as a claim. **In-tree:** POST /api/v1/crucible; 501 CRUCIBLE_NOT_RUNNABLE when no critic dispatchers are composed. Seats in model_rater: crucible.plaintiff, crucible.defense, crucible.judge. Product profile: docs/PROFILES.md. Federated embodiment: Grayson runs Crucible as the **app** on **his** Core (docs/federation/GRAYSON.md); host still NO_HOST until Keith names it.

What it is

Crucible is a **product** on the COSMOS OS. A case (or other legal packet) is evaluated by **named seats** that run **independently first**, then argue. One **disposer** holds the write path. Spend is gated. Critics are a **different family** from the builders and judge the *decision*, not style.

The same occupancy is reused as other skins by changing the **packet**, not by cloning the stack:

Skin	Packet	Seats (embodiments)
Forge	code / work order	CCr + N adversarial coders
Crucible	legal casefile	plaintiff / defense / judge as seats , not as the claimed invention
Diligence	data room / 10-K / deck	bull / bear / independent risk
Differentiator	anonymized clinical case	independent opinions, then argue
Docket	IP packet	applicant / examiner / prior-art

Roles are **embodiments**. The method is: isolate proposers, different-family critics, one disposer, runtime-bind, ledger the round.

How a round runs (enabling)

- 1 Operator (or federated peer) presents a packet and named seats. Each seat is a model id from the local catalog / occupancy file, not a hard-coded vendor.
- 2 **Independent first.** Each proposer receives the packet and the task. They do **not** share a transcript mid-pass. No “second agent based on the first.”
- 3 Artifacts land as proposals (files / job results), not as a merged chat.
- 4 **Argue.** A later stage may exchange the isolated first takes. Disagreement is the signal, not a defect.
- 5 **Critics** (different family) compare the round to the **decided** rubric.
- 6 **One disposer** accepts or refuses. Core ledgers the round. cDeck Review pane paints FINDINGS; it does not invent an approve/reject verb the OS lacks.
- 7 POST /api/v1/crucible without composed critics is **501** - an honest refusal, not a stub job. Fail-closed.

Federation (shape, not a live peer)

COSMOS is the OS. Crucible is the application a peer can run. Grayson: thin Core on **his** PC, **his** keys, **his** tree. Updates/comms from a hub Keith names later (T7 or SRV1 - both NO_HOST today). Mailbox + one versioned API + lease **before** any write that touches Keith’s install. Do not share v:\A. Do not invent a hostname.

cDeck parallel instances (Forge window + Crucible window) are two **clients** of one API, the same sit as Grok.com + OpenWork today. Not two Cores.

What not to claim

Do **not** claim “AI plaintiff, defense, and judge” as the invention. That triad is crowded: SimuCourt/AgentsCourt arXiv:2403.02959; AgentCourt arXiv:2408.08089; **CN119168059B granted** 2025-07-15; **US20260037351A1** claim 21; jury-side US20250148558A1. File Crucible only as **P05 occupancy applied to a legal packet**. Else

fold into P05 and spend the 12th slot on UPS-JUDGE with Keith's July pack.

Name collisions (not TESS): CrucibleTech RN 8043152; Star Lab CRUCIBLE RN 5023199; Crucible Discovery; Destiny PvP fame.

Already public (clock)

GitHub bts-mesh **About** text names "CRUCIBLE method" as of 2026-08-16T10:21:14Z. Tracked files in that repo: **zero** hits for CRUCIBLE. Public date is a **name**, not a method disclosure in code. US grace for that string is thin.

Attach to

P05 (parent occupancy) at minimum. P12 only if Legal files occupancy-not-roles. Do not attach this appendix as if it cured a PDJ claim.

Appendix - How BTS-MESH works (July 2026 predecessor)

Kind: written description / provenance for Legal to attach to US provisionals. **Status:** not a USPTO filing. Excerpts from the local predecessor tree `V:\Ai\BTS_MESH` (GrokBot pen). This TUI **read** those files; it does **not** write that tree. Do not commit keys, OAuth, or live ledgers. **Purpose:** prove the mesh **ran in early/mid July 2026**, and describe the architecture COSMOS later replaced - so provisionals are not a history rewrite.

Dates (measured)

Fact	Date	Where
North-star architecture written	2026-07-10	BTS_MESH_NORTH_STAR.md (local)
Working foundation README	2026-07-10 / README stamp 2026-07-11	README.md - bus, registry, cards, watcher, orchestrator; 29 tests
SOP / wiring / state feed	2026-07-14	BTS_SOP.md, WIRING.md, BTS_STATE_2026-07-14.md
GitHub bts-mesh created	2026-08-16T10:21:14Z	public repo; About names CRUCIBLE; tracked files have 0 CRUCIBLE hits
COSMOS architecture ratified	2026-08-23	GitHub cosmos; docs/RATIFIED.md

July operation = **local run records**. GitHub August 16 is **not** the start of operation. Private-now does not un-publish GitHub.

The goal (Keith, recorded 2026-07-10)

Grok could start a new session of another model (or ChatGPT) and load/point to needed context, and vice-versa, as new sessions and topics spawn - multi-session, multi-agent, multi-AI simultaneous coordination - and get Keith out of the middle as the human message-wire.

The one insight: no consumer chat tab can autonomously boot **and seed** another vendor's session. That capability has to live in a **local orchestrator** on the operator's machine.

Five components (as of 2026-07-10, working code)

- 1 **Shared bus** (`bts_bus.py`). Append-only, author-prefixed signal log. Payloads written atomically (temp + rename) so no reader sees a half-file. Per-reader cursor; self-filtering so a side never eats its own signals.
- 2 **Session seed**. Every spawned session starts by reading a `SESSION_SEED-<topic>.md` pointer. "Load the needed context" = read the seed. Orchestrator auto-creates seeds from a template. This is BootUP generalized.
- 3 **Session registry** (`SESSION_REGISTRY.json`). Single source of truth for {agent, topic, url, status, tasks}. Prevents two sessions doing the same work.
- 4 **Capability cards + routing**. Each AI advertises what it is best at; the daemon routes a new topic to a node. Cards are data (`cards/*.json`).
- 5 **Orchestrator daemon** (`bts_orchestrator.py`). Consumes the bus, updates the registry, handles `SPAWN_SESSION` (route to seed to register to queue launch). `launch_session()` was the remaining stub (Chrome bridge).

Wire format (contract v1.1): {type, path, note, ts} plus routing extras. Types included `NEW_OUTPUT_READY`, `QUESTION_FOR_YOU`, `NEEDS_INPUT`, `TASK_COMPLETE`, `ACK`, `SPAWN_SESSION`.

Honest ceilings recorded in July (not smoothed)

- Not real-time. File-paced (seconds to minutes).
- Browser legs brittle. Prefer vendor APIs; fall back to DOM.
- More nodes = more drift (fake-DONE / contradictory state). Mitigations named

then: append-only log, single-source registry, adversarial cross-check. Keep topology supervisor + fan-out, not all-to-all.

What COSMOS kept, and what it replaced

BTS-MESH is the **predecessor mesh**. COSMOS is the **OS** that took the same job - multi-AI coordination without Keith as the wire - and put **authority** in one resident Core:

BTS-MESH (July)	COSMOS (ratified 2026-08-23)
File bus + registry as coordination	One signed JSONL ledger as authority; projections rebuild
Orchestrator daemon as conductor	Core = API + scheduler + arbiter + sole ledger writer
Atomic temp+rename payloads	Fenced commit gateway + fencing tokens
Session seed files	Signed SEED . json at session close (OPEN_CONTEXT if missing)
Capability cards	Registry + rails matrix; DOM-first policy
Keith still in some loops	CCr one writer; orch does not hold the COSMOS pen

Continuity of **operation** (July run to August GitHub to live Core) is provenance. Continuity of **code** is successor architecture, not a claim that the July bus is the 2026-08-23 ledger.

What this is not

Not a dump of the BTS tree into a patent. Not keys, Drive folder IDs, or OAuth. Not a claim that GitHub bts-mesh disclosed the method in tracked files (it did not, for CRUCIBLE). Not permission to flip GitHub private from this TUI.

Attach to

P07 (Core as successor authority), P05 (occupancy / multi-AI coordination), P11 (seed / carry-over), and the provenance note Legal wants on the public record. Duplicate into each FILE provisional that needs July operation as support for 112(a) or for inventor-grace timing. Legal copies further excerpts from v:\Ai\BTS_MESH; this TUI does not.